

Carrier-Frequency Dependence of a Step-Modulated Pulse Propagating Through a Single Narrow-Resonance Absorber

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Abstract

We study optical precursors when a step-modulated pulse propagates through collections of cold potassium atoms. Here, we expand on our previous report on the direct observation of optical precursors [PRL **96**, 143901, (2006)], giving more experimental details and extending our work to the case of an off-resonance carrier frequency. The transmitted pulse consists of a several-nanosecond-long spike occurring immediately after the front. The spike begins with nearly 100% transmission, consisting of both the Sommerfeld and Brillouin precursors, and subsequently decays to a constant value expected from Beer's Law of absorption. For the case of an off-resonance carrier frequency, we observe interference between the optical precursors and the main signal, causing modulation due to the detuning of the carrier frequency from the medium resonance frequency. Precursors may be useful for imaging applications requiring penetrating optical radiation, such as in biological systems, or in optical communication systems.

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I. INTRODUCTION

Many modern optical technologies, such as optical communication and medical imaging, require an understanding of how optical pulses propagate through a dispersive medium. Recently, the advent of methods for tailoring the dispersion of materials enables exquisite control of pulse propagation characteristics. For example, it is now possible to achieve an exceedingly slow or fast group velocity v_g compared to the speed of light in vacuum c for pulses of light propagating through a gas of atoms [1]. The achievement of slow and fast light has triggered fundamental questions about the information velocity as it relates to causality in Einstein's special theory of relativity. Over the past decade, researchers have proposed that encoding information on a pulse is related to non-analytic points. Because it is predicted that non-analytic points travel at c , it follows that the information velocity is also equal to c [2–4]. Stenner *et al.* confirmed this in both slow- [5] and fast-light experiments [6], where they encoded one bit of information on a pulse by rapidly changing the pulse intensity. By tracking when it is first possible to identify the bit, they showed that the information velocity is equal to c regardless of the dispersive properties of the medium.

It is generally believed that the rapid change of intensity on a pulse creates optical precursors [7–10], the transient behavior of the propagated field. In the experiment by Stenner *et al.*, however, it was difficult to distinguish the precursors from the rest of the pulse because the rapid change in pulse intensity occurred in the middle of a time-dependent pulse waveform. To address this shortcoming, experiments need to be conducted that clearly tests for the existence of optical precursors occurring when a step-modulated laser pulse propagates through a linear dispersive medium. We reported our preliminary work on the first direct observation of long-lived optical precursors in Ref. [11] for the case when the carrier frequency of the pulse ω_c was equal to the material resonance frequency ω_0 . The measurement was facilitated by extending the time scale of the precursors up to the order of tens of nanoseconds using a narrow-resonance absorber consisting of cold potassium atoms contained in a magneto-optical trap (MOT). Recently, optical precursors were observed experimentally in a medium rendered transparent through electromagnetically-induced transparency (EIT) [12].

In this paper, we extend our previous work [11] to discuss details of the experiments and the results, especially for the case of off-resonance excitation ($\omega_c \neq \omega_0$). In Sec. III,

we describe the experimental method used to increase the duration of optical precursor, which allows us to detect them directly using a standard experimental apparatus. The experimental results are described in Sec. IV. In addition to the longer duration, we observe a large precursor amplitude for the case of an on-resonance carrier frequency, which is comparable to the size of the main signal. For the off-resonance case, we observe that the frequency difference between optical precursors and main signal generates oscillatory behavior. In Secs. V A and V B, we compare our experimental observations to two different theories. One is based on an asymptotic analysis of the optical wave equation and a classical model of the dielectric, where the asymptotic analysis takes advantage of the fact that the material is weakly dispersive and has a narrow resonance [11]. The other is based on an analytic solution of the pulse propagation equations under the narrow-resonance, dilute-gas approximation, an approach commonly used by the quantum optics community [13]. From these analysis, we identify features of the experimentally observed propagated field that correspond to the precursors.

II. HISTORICAL BACKGROUND OF OPTICAL PRECURSORS

Optical precursors were first introduced by Sommerfeld and Brillouin in the early 1900's to demonstrate that signals cannot travel faster than c even in regions of anomalous dispersion for which $v_g > c$ or $v_g < 0$ [7]. To characterize the signal propagation, they considered a step-modulated pulse, which is turned on suddenly at a time before which there is no field. According to their analysis, detecting precursors is difficult because of their small intensities compared to the main signal. The Sommerfeld (Brillouin) precursor is predicted to have a maximum intensity of 10^{-7} (10^{-4}) of the eventual main signal intensity. Although Brillouin's analysis is quoted in many electricity and magnetism textbooks [8, 9], several researchers have identified mistakes in his early work, as reviewed by Oughstun and Sherman [10]. Most notable is the modern prediction that the precursors are much larger than previous thought; their amplitude can be similar to or larger than the amplitude of the main signal. Even though the precursors are predicted to be large, the conventional wisdom is that they are a strictly an ultra-fast phenomena, lasting for only a few optical cycles and hence indirect detection methods are required. This conclusion, however, is based on the fact that the theories have only been evaluated for a medium with a broad resonance, as discussed below.

More recently, several methods have been proposed to observe precursors in specific systems based on numerical analysis of the matter-field equations. These include the study of Brillouin precursors for a step-modulated microwave field propagating through a biological sample (mostly water) [14], Brillouin precursor propagation in ultra fast optics [15], and Sommerfeld precursor propagation in photonic crystals [16]. However, none of these previous studies have made a quantitative comparison with the asymptotic theory developed by Oughstun and Sherman [10] for the case of pulse propagation through a resonant dielectric in a region of anomalous dispersion.

Contrary to the theoretical developments, whose history is given in Ref. [10], there are only a few experimental demonstrations of optical precursors. Precursor-like behavior was first observed for microwave pulse propagation in a ferrimagnetic coaxial waveguide near cut-off [17] based on the Brillouin's asymptotic theory [7]. This experiment used artificially created dispersion characteristics and hence did not test the theories of pulse propagation through a Lorentz dielectric.

By virtue of modern experimental technologies, precursors have been observed indirectly in the optical domain [18–20] and observed in the acoustic domain as hydrodynamic transient surface wave phenomena in a thin layer of mercury [21]. Aaviksoo *et al.* [18] indirectly observed optical precursors by propagating an ultra fast single-sided exponential pulse through thin GaAs semiconductor crystals. The fast propagating front of the pulse was observed regardless of the carrier frequency detuning from the excitonic resonance frequency. The main signal, which follows after the fast front, experienced a detuning-dependent delay. Notably, they used the so-called slowly varying amplitude (SVA) approximation to explain theoretically the observed temporal evolution of the total intensity transmission, but never compared this theory to the asymptotic theory, which is needed to correctly identify the precursors. Sakai *et al.* [19] also reported the observation of optical precursors when they studied polariton pulse propagation at an exciton resonance. They sent a sub-picosecond pulse through a semiconductor (CuCl) when the carrier frequency of the pulse was set to the excitonic resonance. They interpreted the pulse front as an optical precursor because the front consists of low frequency component. However, they used a rather high pulse intensity and therefore could not unambiguously rule out the possibility of nonlinear phenomena affecting the propagated pulse. An observation of optical precursors in pure water was reported by Choi *et al.* [20]. They propagated 540 fs-long pulses (carrier frequency is off-resonance from

the resonances of water) through 700 mm of deionized water and observed pulse breakup. Interpretation of their results created some controversy regarding the non-exponential decay of optical precursors [22, 23] and the existence of optical precursors [24]. Regardless of the debates, all the previous experiments have only considered broad resonances.

Using narrow resonances, as reported in Ref. [11], we observed directly optical precursors created when a step-modulated field propagated through a collection of cold potassium atoms that has a narrow resonance. The existence of Sommerfeld-Brillouin precursors in the data was theoretically confirmed based on an asymptotic analysis by LeFew *et al.* [25]. After that, followup research included the study of the main signal and precursors for a Gaussian-modulated input pulse [26], and the investigation of precursor behavior when the system parameters are varied from narrow-resonance to broad-resonance parameter regimes [27]. More recently, it was realized that the spikes occurring in the biphoton waveform generated by an electromagnetically-induced transparency (EIT) medium are a manifestation of optical precursors [28]. Optical precursors were also investigated for a classical step-modulated pulse propagating through an EIT medium both theoretically [29, 30] and experimentally [12]. Most recently, the observability of coherent stacks [29] of optical precursors generating 700% transmission was both theoretically proposed [31] and experimentally observed [32].

Based on our work [11], we believe that there have been several experiments that have observed precursors for the case of a weakly-dispersive narrow resonance medium [33, 34], but this realization was unrecognized. A reason was the different time scale of these experiments compared to the femtosecond time scale expected for precursors based on a theory that considers a broad material resonance. In this regard, Crisp [13] states, without proof, that the SVA approximations rules out precursor phenomena because the envelope of the field must vary slowly. Thus, this well known approach of using the SVA in the quantum optics community [35] has not been connected to precursor phenomena. However, the previous observations of “coherent transients” [33, 34] has recently been identified as precursors [25, 36]. For example, opposed to Crisp’s comments, several researchers have suggested that the SVA theory is applicable to precursor phenomena. Varoquaux *et al.* [37], for example, noticed that “acoustic wave propagation in superfluid $^3\text{He-B}$ ” [38] and Rothenberg’s observation of a “weak 0π pulse” [39] could be a “0th order precursor” through the use of a SVA analysis. Aaviksoo *et al.* [40] also applied the SVA theory to their observation [18], but they could not distinguish between Sommerfeld and Brillouin precursors. Neither group of

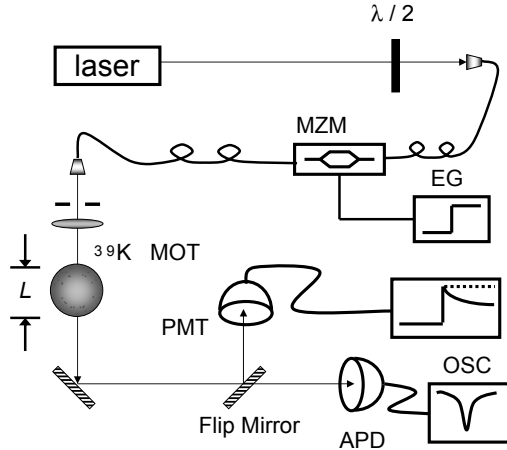


FIG. 1: Experimental setup. EG: Edge generator, OSC: oscilloscope, APD: avalanche photodiode, PMT: photo multiplier tube, MZM: Mach-Zehnder modulator.

researchers compared the coherent transients to the predictions of the asymptotic theory used by Sommerfeld and Brillouin until recently [25].

III. EXPERIMENTS

To directly measure optical precursors, three steps are used. First, we prepare a cloud of cold potassium (^{39}K) atoms to obtain a dielectric medium with a narrow resonance, which is the key to extending the precursor duration [Stage 1 in Fig. 2]. The cold atoms are optically pumped into one of the ground states of the potassium $4S_{1/2}$ state ($F = 1$), thereby preparing essentially a single-resonance Lorentz dielectric [Stage 2 in Fig. 2]. Once the atoms are optically pumped, we send a weak-intensity, step-modulated pulse (carrier frequency ω_c) through the medium and measure the intensity of the transmitted pulse. In this section, we give further details of each experimental step.

A. Stage 1: Preparation of a narrow resonance medium

To achieve a narrow-resonance, we use a vapor-cell magneto-optic trap [41–44]. The magneto optic-trap for neutral potassium atoms (^{39}K) is realized using laser beams tuned to the $4S_{1/2} \leftrightarrow 4P_{3/2}$ transition (D_2 transition, 767-nm transition wavelength) and anti-Helmholtz coils. The trapped atoms have a temperature of $\sim 400 \mu\text{K}$ measured by the release-and-recapture method [45]. At such cold temperatures, the resonance linewidth becomes of-

the-order-of the natural linewidth (full width at half maximum) $2\delta/2\pi = 1/2\pi t_{sp} \sim 6$ MHz, where t_{sp} is spontaneous decay time of ^{39}K . This width is much narrower than the resonance width of a Doppler-broadened potassium vapor at room temperature, which is typically 800 MHz. The diameter of the MOT, which we take to be the length L of the medium, is determined to be in the range of ~ 1 -2 mm by measuring the $1/e$ of the fluorescence of the MOT. The atomic number density is ~ 1 - $2 \times 10^{10} \text{ cm}^{-3}$ by measuring the absorption of a weak continuous-wave probe beam passing through the MOT.

The optical precursor experiment is conducted on the $4S_{1/2} \leftrightarrow 4P_{1/2}$ transition (D_1 transition, 770-nm transition wavelength), as shown in Fig. 2 (c)-(d). At the D_1 transition, there are two ground states ($F=1,2$) and two excited states ($F'=1,2$). The corresponding four resonance peaks are denoted as $\omega_{F'F}$, where $F'F$ denotes the $4S_{1/2}(F) \leftrightarrow 4P_{1/2}(F')$ transition, as seen in Fig. 2 (c). At Stage 1, the two MOT beams (tuned near the D_2 transition) are on, and the ground states are equally populated so that the absorption from the ground states to the excited states are well balanced, as shown in Fig. 2 (a), (c), and (e). As discussed in the next section, ω_{21} will be treated as a single resonance peak. Therefore, we let the resonance frequency of the single-Lorentz medium $\omega_0 \equiv \omega_{21}$ in this paper.

B. Stage 2: Preparation of a single-Lorentz dielectric

To obtain a single-resonance Lorentz medium, we optically pump the atoms into one of the ground states $4S_{1/2}(F=1)$. To achieve optical pumping, the repumping beam (red detuned from $4S_{1/2}(F=1) \leftrightarrow 4P_{3/2}$ transition) is repeatedly switched off for 20 μs so that the remaining trapping beam optically pumps the atoms out of the $F=2$ state and into the $F=1$ state [Fig. 2(f)]. The repumping beam is then turned back on for 80 μs , returning to Stage 1 [Fig. 2(e)]. Once in this state, the optical precursor experiment is conducted and the process is repeated.

The optical pumping time interval of 20 μs is chosen to balance two conditions. One is to have enough time for the atomic states to reach equilibrium after optical pumping and to perform the optical precursor experiment. At the same time, we need to keep the total number of atoms constant in the trap during the optical pumping time.

After optical pumping, approximately 88% of the atoms are in the $F=1$ state, leading to a suppression of resonances ω_{12} and ω_{22} , and enhancement of resonances ω_{11} and $\omega_0 = \omega_{21}$,

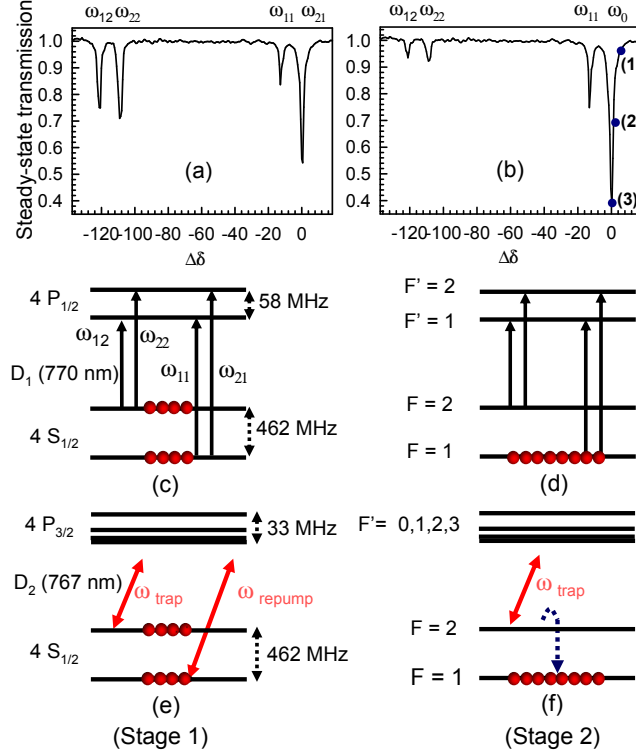


FIG. 2: Transmission of a weak probe beam whose frequency is scanned through the four resonance peaks of the potassium D₁ transition for the case of (a) both MOT beams on for 80 μ s, and (b) optical pumping into F=1 state with trapping beam off for 20 μ s. Energy level diagrams of the D₁ transition and population distribution in the ground states are shown in (c) and (d), explaining the origin of the four resonances shown in (a) and the two resonances shown in (b), respectively. Energy level diagrams of the D₂ transition and population distribution in the ground states, are shown in (e) and (f) to illustrate the process of optical pumping.

as seen in Fig. 2 (b). The difference in the heights of resonances ω_{11} and ω_0 is due to differences in the corresponding dipole matrix elements. Resonance ω_0 is well isolated from resonance ω_{11} (its width is much narrower than the spacing between the resonances.) and we will use it to approximate a single-resonance Lorentz dielectric.

The properties of a single-resonance Lorentz dielectric are characterized by measuring the resonance linewidth δ and the line-center absorption coefficient α_0 of the absorption peak near $\omega_{21} = \omega_0$, as shown in Fig. 2(b). From the measurements of the frequency-dependent steady-state probe-laser-beam transmission T at line center [Fig. 2 (b)], we determine half-width at half-maximum δ and line-center absorption coefficient α_0 . From the measured

values, we determine α_0 through Beer's Law $T = \exp[-\alpha_0 L]$, and a Lorentzian-like resonance with a width $2\delta/2\pi \sim 9.6$ MHz (full width half maximum), which is broader than the 6-MHz natural linewidth. The broadened linewidth consists of residual Doppler broadening (<1 MHz), Zeeman splitting arising from the magnetic field gradient (~ 2 -3 MHz), and the laser linewidth (~ 200 kHz). Because the time scale of precursor decay is of-the-order-of $1/2\delta$, as we discuss later, the estimated precursor time scale is $1/2\delta \sim 26$ ns.

Experimental measurements are obtained for a single value of the on-resonance absorption path length, set by adjusting the intensities of cooling and trapping laser beams. At their maximum intensity (~ 33 mW/cm²), we achieve an atomic number density of 1.2×10^{10} cm⁻³ and a propagation length $L=0.20$ cm, resulting in a transmission of $T=0.36$, corresponding to $\alpha_0 = 5.14$ cm⁻¹, $\alpha_0 L = 1.03$. With the trapping beams off, there are no trapped atoms and the transmitted pulse is identical to the incident pulse, which serves as a reference.

C. Stage 3: Step modulated pulse propagation

The next stage is to pass a step-modulated pulse through the medium, and measure the temporal evolution of transmitted pulse intensity $T(z, t)$, where the z is the medium depth, and $\tau = t - z/c$ is the retarded time. The incident pulse is created by passing a weak continuous-wave laser (frequency ω_c) through a 20-GHz-bandwidth Mach-Zehnder modulator (MZM, EOSpace, Inc.) driven by an electronic edge generator (Data Dynamics, Model 5113) whose edge is steepened using a back recovery diode (Stanford Research Systems, Model DG535), resulting in an edge with a rise time of ~ 100 -200 ps. The peak intensity of the pulse is ~ 64 μ W/cm², which is much less than the saturation intensity of the transition (~ 3 mW/cm²). A weak-intensity incident pulse is important to avoid nonlinear optical phenomenon. The pulse is detected with a fast-rise-time (0.78 ns) photomultiplier tube (PMT, Hamamatsu, Model H6780-20) and the resulting electrical signal is measured with a 1-GHz-analog-bandwidth digital oscilloscope (Tektronix, Model TDS 680B). In the absence of the atoms, we measure an edge rise time (10%-90%) of ~ 1.7 ns for the complete system, corresponding to a bandwidth of ~ 206 MHz. This time is short in comparison to the expected duration of the precursors.

To calibrate ω_c , we measure its detuning with respect to the atomic resonance frequency via the steady-state transmission spectrum. From knowledge of the hyperfine splitting of

the $4P_{1/2}$ state (58 MHz [46]), we calibrate the horizontal axis of the scan. The detuning $\Delta = \omega_c - \omega_0$ is then determined by comparing the steady-state transmission spectrum with the value of the long-time intensity of the transmitted pulse in our transient experiments with values $\Delta_{(1)} = 27.1^{+3.2}_{-2.1}$ MHz, $\Delta_{(2)} = 5.3^{+0.5}_{-0.4}$ MHz, and $\Delta_{(3)} = 0^{+2.3}_{-2.3}$ MHz indicated in Fig. 2. The errors associated with these measurements arise from our mapping of transmission to frequency and are asymmetric due to the fact that the slope of the curve is different at each point.

IV. EXPERIMENTAL RESULTS

In this section, we describe our experimental findings in which we measure $T(z, \tau)$ for different carrier frequencies ω_c , including the on-resonance case ($\omega_c = \omega_0$). The black solid lines in Fig. 3 show the measured transmitted pulse intensities for different carrier frequencies tuned near the $4S_{1/2}(F=1) \leftrightarrow 4P_{1/2}(F=2)$ transition.

Figure 3 (a) [point (1) in Fig. 2(b)] shows the case when $\Delta = \Delta_{(1)} \sim 5\delta$. It is seen that the transient transmitted intensity immediately reaches $\sim 90\%$ of the incident pulse height. The transmitted intensity is ideally 100%, but it is reduced to 90% by the 206-MHz electronics bandwidth. The transmission intensity oscillates with a modulation frequency of approximately Δ and a modulation period of 40 ns. The amplitude of oscillations then decays to the steady-state value, which obeys Beer's law. The time scale for the transmitted intensity to reach its steady-state value is similar to that observed for $\Delta = 0$, as shown in Fig. 3 (c), which implies that the precursor time scale only depends on δ . As discussed later, the oscillation of the envelope results from beats between the precursors oscillating at ω_0 and the main signal at ω_c . For the case of a moderately blue-shifted carrier frequency ($\Delta = \Delta_{(2)} \sim \delta$), as shown in Fig. 3 (b) [point (2) in Fig. 2 (b)], the initial transmission also rises immediately to $\sim 90\%$, decays to the steady state with slower oscillation than the case for $\Delta = \Delta_{(1)} \sim 5\delta$. Note that, for the case of $\Delta = \Delta_{(1)} \sim 5\delta$, shown in Fig. 3 (a), $T(z, \tau)$ oscillates after the peak until it reaches its steady-state value. During the oscillation, it attains values greater than unity. This phenomena can be explained as follows. When the medium polarization starts to react to the incident light, the polarization is out of phase with the incident field, indicating that the energy of incident light is stored temporarily by the medium. The stored energy is re-emitted afterward at a later time, causing greater than

unity transmission.

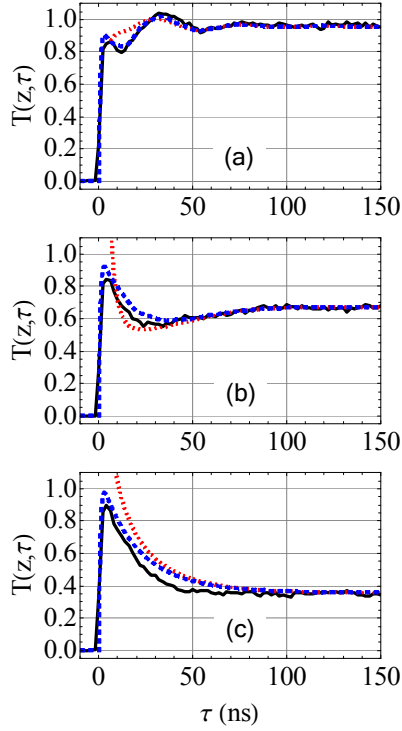


FIG. 3: Experimentally obtained transient transmission intensity (black solid lines) compared with two theoretical analysis: the asymptotic analysis (Eqs. (8)-(9) and Eq. (10), red dotted lines), and the weakly dispersive narrow resonance (Eqs. (13)-(18) and Eq. (18), blue dashed lines). Transient transmission taken near the $4S_{1/2}(F=1) \leftrightarrow 4P_{1/2}(F=2)$ transition for (a) $\Delta = \Delta_{(1)} \sim 5\delta$, (b) $\Delta = \Delta_{(2)} \sim \delta$, and (c) $\Delta = \Delta_{(3)} \sim 0$.

V. ANALYSIS: TWO APPROACHES

The experimental results described in the previous section clearly demonstrate transient behavior, yet it is not clear which part of the waveform can be attributed to the precursor or to the main signal parts of the field.

Consider a dispersive medium consisting of a collection of Lorentz oscillators possessing a single resonance. The refractive index for this medium (or a collection of two-level atoms) is given by

$$n(\omega) = \sqrt{\epsilon(\omega)} = \sqrt{1 - \frac{\omega_p^2}{\omega^2 - \omega_0^2 + 2i\omega\delta}}, \quad (1)$$

where $\epsilon(\omega)$ is linear frequency-dependent dielectric constant, ω_p is the plasma frequency, δ is the resonance absorption linewidth (HWHM), and ω_0 is the atomic resonance frequency. The plasma frequency ω_p quantifies the strength of the resonance and is related to the line-center absorption coefficient through the relation $\alpha_0 \simeq \omega_p^2/2\delta c$, which is only true for small ω_p . The temporal evolution of the transmitted scalar electric field $E(z, t)$ at a medium penetration depth z in the half-space $z > 0$ can be written as an integral representation given by

$$E(z, t) = \frac{1}{2\pi} \int_{ia-\infty}^{ia+\infty} E(0, \omega) e^{i(k(\omega)z - \omega t)} d\omega, \quad (2)$$

where a is a positive definite real constant and

$$E(0, \omega) = \int_{-\infty}^{+\infty} E(z = 0_+, t) e^{i\omega t} dt \quad (3)$$

is the temporal Fourier spectrum of the incident beam just inside the dispersive medium. The complex wave number $k(\omega)$ is related to the complex index of refraction through $k(\omega) = \omega n(\omega)/c$. If the incident beam is taken as a step-modulated sinusoidal electric field of the form $E(z = 0, t) = \text{Re}[E_0 \Theta(t) e^{-i\omega_c t}] = E_0 \Theta(t) \sin(\omega_c t)$, where $\Theta(t)$ is the Heaviside unit step function with spectrum i/ω for $\text{Im}\{\omega\} > 0$, then the transmitted field is given by

$$E(z, t) = -\frac{E_0}{2\pi} \text{Im} \left\{ \int_{ia-\infty}^{ia+\infty} \frac{i}{\omega - \omega_c} e^{z\phi(\omega, t)/c} d\omega \right\}. \quad (4)$$

Here, the complex function $\phi(\omega, t)$ appearing in Eq. (4) is defined as $\phi(\omega, t) \equiv i\omega[n(\omega) - ct/z]$. Equation (4) is the starting point of the theories.

The transmitted field, expressed in integral form in Eq. (4), consists of two parts: transient responses (the precursors) and steady-state responses (the main signal). However, this equation has no exact analytic solution. To identify each part, we will discuss two theoretical approaches to solve Eq. (4) and compare their predictions to our data.

A. The numerical asymptotic method

Equation (4) for the total transmitted field can be evaluated using the saddle-point method, which is valid in the limit when a distance into the medium, z , is greater than one optical penetration depth α_0^{-1} [10, 47]. In the asymptotic regime, the integral has a non-zero value when it is evaluated near the extremum value (saddle-points) of the phase

$z\phi(\omega, t)/c$. The extremum values are the so-called saddle-points ω_{sp} , which are solutions to $\partial_\omega \phi(\omega, t)|_{\omega_{sp}} = 0$. At each ω_{sp} , the contribution to integral Eq. (4) is given by

$$E_{\omega_{sp}}(z, t) = -\text{Im} \left\{ \frac{iE_0}{\sqrt{2\pi}(\omega_{sp} - \omega_c)} \frac{e^{z\phi(\omega_{sp}, t)/c + i\psi}}{|z\phi''(\omega_{sp})/c|} \right\}, \quad (5)$$

where ψ is the angle of steepest decent [48].

In our case of a weakly-dispersive narrow-resonance medium and $\Delta = 0$, the two classes of saddle points $\omega_{sp}^\pm$ are evaluated (Ref. [12, 25]) as

$$\omega_{sp}^\pm = \omega_0 - i\delta \pm \sqrt{p/\tau}, \quad (6)$$

where $p \equiv \omega_p^2 z / 4c = \alpha_0 \delta z / 2$. These saddle points are related to the two types of transient responses known as Sommerfeld [$E_S(z, t)$] and Brillouin precursors [$E_B(z, t)$] as appearing in Eqs. (5)-(6) of Ref. [30]. The total Sommerfeld-Brillouin precursors, $E_{SB}(z, t) = E_S(z, t) + E_B(z, t)$, takes the form of a modulated cosine or Bessel function [11], as obtained analytically in recent papers [12, 25, 31].

To make predictions about the precursors for $\Delta = 0$ in our previous work [11], we used Oughstun and Sherman's asymptotic theory [10]. Unfortunately, approximations made in their theory resulted in a large error of about 10^{10} near $\tau = 0$ because the approximations are only valid for a material that is strongly dispersive ($\omega_p \sim \omega_0$), has a broad resonances ($\delta \sim 0.1\omega_0$), and far off-resonant excitation ($\omega_c \sim 0.25\omega_0$). The error was corrected by recent theoretical work on asymptotic theory by LeFew *et al.* [25]. The asymptotic theory is valid for the optically thick regime: $\alpha_0 L \gg 1$ [12, 25, 28–30].

Section IV. of Ref. [25] discusses the range of validity of the asymptotic analysis, which depends on the material parameters and τ . For our experimental parameters, the analysis is accurate when $\tau \gg 64.7$ ns according to Eq. (59) of Ref. [25]. Therefore, the analytic expression of the asymptotic theory can be used for small $\alpha_0 L$ [25] and has confirmed our interpretation that the transients observed in our experiment consist of Sommerfeld and Brillouin precursors for the case $\Delta = 0$ [11].

To investigate the off-resonance case $\Delta \neq 0$, we determine the location of the saddle-points and the amplitudes of each precursor numerically, because saddle points obtained analytically [Eq. (6)] are suitable only for the case when $\Delta = 0$ [25].

As discussed by LeFew *et al.* [49], we set the variable of integration as $\xi \equiv i(\omega + i\delta)$, which is shifted by $-\delta$ from the imaginary axis of complex frequency ω and rotated by 90

degrees. The phase of the integrand is thus simplified as $z\phi(\omega, t)/c \equiv z(\xi + \delta)[R_1/R_2 - \theta]/z_0$, and the saddle-point equation is now given by

$$R_1^3 R_2 \theta - R_1^2 R_2^2 + (R_2^2 - R_1^2) \xi(\xi + \delta)|_{\xi_{sp}} = 0, \quad (7)$$

where $z_0 \equiv \omega_0/c$, $R_1 \equiv \sqrt{(\xi^2 + \omega_0^2 - \delta^2)/\omega_0^2}$, and $R_2 \equiv \sqrt{(\xi^2 + \omega_0^2 + \omega_p^2 - \delta^2)/\omega_0^2}$. Using the numerically obtained values of four saddle points $\xi_{sp}^\pm$ ($\xi_S^\pm$ and $\xi_B^\pm$), we then evaluate the non-vanishing field envelope $A(z, t)$ of each precursor using Eq. (5) as

$$A_S(z, t) = \sum_{\xi_S^\pm} \frac{\mp E_0}{\sqrt{2\pi}} \frac{e^{\frac{z}{z_0} \phi(\xi_S^\pm(\theta), \theta) - \frac{i}{2} \text{Arg}[\phi_{\xi\xi}(\xi_S^\pm(\theta))]} }{(\xi_S^\pm(\theta) + \delta - i\omega_c) \sqrt{\frac{z}{z_0} |\phi_{\xi\xi}(\xi_S^\pm(\theta))|}}, \quad (8)$$

$$A_B(z, t) = \sum_{\xi_B^\pm} \frac{\mp E_0}{\sqrt{2\pi}} \frac{e^{\frac{z}{z_0} \phi(\xi_B^\pm(\theta), \theta) - \frac{i}{2} \text{Arg}[\phi_{\xi\xi}(\xi_B^\pm(\theta))]} }{(\xi_B^\pm(\theta) + \delta - i\omega_c) \sqrt{\frac{z}{z_0} |\phi_{\xi\xi}(\xi_B^\pm(\theta))|}}, \quad (9)$$

where $\sum_{\xi^\pm}$ indicates for the summation over each case of ξ^+ on the $\text{Im}(\xi) > 0$ plane and ξ^- on the $\text{Im}(\xi) < 0$ plane.

Figure 4 shows the Sommerfeld precursor [Eq. (8)] and the Brillouin precursor [Eq. (9)] separately. Blue lines denotes the case of $|\Delta| \sim \delta$, and the red lines indicate the case of $|\Delta| \sim 5\delta$. For $\Delta = 0$, both precursors have the same amplitude [black lines]. When the carrier frequency ω_c is set higher than the medium resonance ω_0 , i.e. $\Delta > 0$ [Fig. 4 (a)-(b)], the Sommerfeld precursor (high frequency transient) transmits through medium with larger amplitude than the amplitude of the Brillouin precursor. For $\Delta > 0$, on the other hand, the Brillouin precursor (the low frequency transient) dominates over the Sommerfeld precursor as shown in Figs. 4 (c)-(d). The numerically obtained total precursor amplitudes $|A_{SB}(z, t)/E_0|$ are shown in Fig. 5 (e) for different Δ .

The saddle points are not the only source of a non-zero contribution to the integral. Note that the saddle points contribute the most when $1/(\omega - \omega_c)$ varies slowly compared to $\exp[z\phi(\omega, t)/c]$. For $\omega = \omega_c$, however, $1/(\omega - \omega_c)$ has a singular point (pole), where it diverges rapidly. The pole contribution to the integral is related to the steady-state response of the medium to the incident field. The steady-state response is known as the main signal [See Fig. 5 (b)],

$$A_C(z, t) = 2\pi i \text{Res}(\xi = i\omega_c - \delta), \quad (10)$$

which is identical to the steady-state term predicted by the analytic expression for the weakly dispersive narrow-resonance case discussed in the next section.

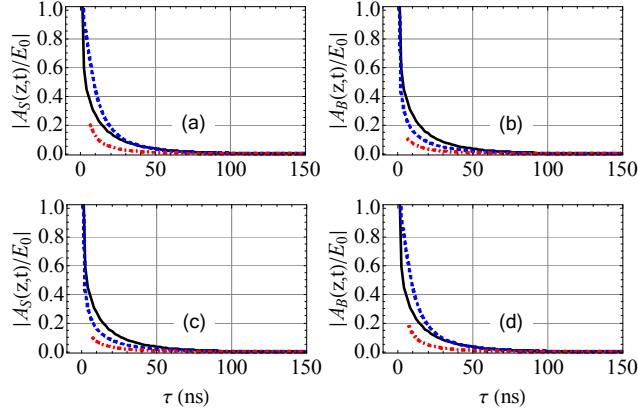


FIG. 4: Value of the field envelopes obtained by using the numerical asymptotic theory given by Eqs. (8)-(9). The Sommerfeld precursors $|A_S(z, t)/E_0|$ are evaluated for (a) $\Delta > 0$, and (c) $\Delta < 0$ and the Brillouin precursors $|A_B(z, t)/E_0|$ are evaluated for (b) $\Delta > 0$ (d) $\Delta < 0$. In each figure, there are three cases for $\Delta = 0$ (black solid lines), $|\Delta| \sim \delta$ (blue dashed lines), and $|\Delta| \sim 5\delta$ (red dash-dot lines).

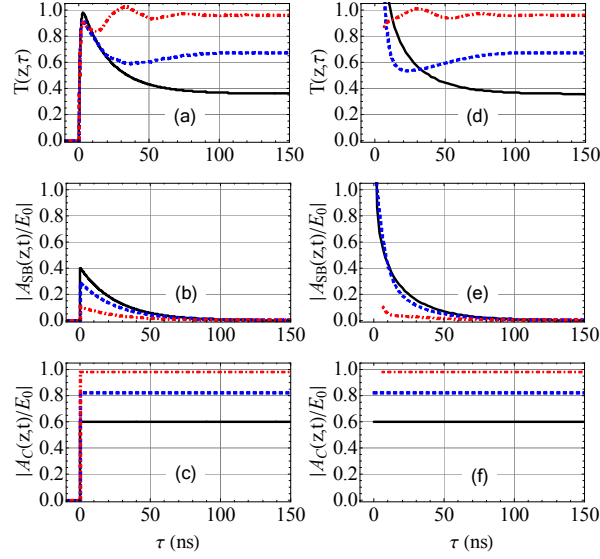


FIG. 5: Comparison of two theories: the weakly dispersive narrow resonance case [(a)-(c), Eqs. (13)-(15)] and asymptotic theory [(d)-(f), Eqs.(8)-(9) and Eq.(10)]. The total transmission $T(z, t)$ [(a), (d)], envelopes of total precursors $|A_{SB}(z, t)/E_0|$ [(b), (e)], and main signal $|A_C(z, t)/E_0|$ [(c), (f)] are evaluated for our experimental conditions. In each figure, the red dash-dot line denotes $\Delta \sim 5\delta$, the blue dashed line indicates $\Delta \sim \delta$, and the black solid line denotes $\Delta \sim 0$.

B. Analytic expression for weakly dispersive, narrow-resonance case

In the previous section, for the off-resonance cases, individual Sommerfeld and Brillouin precursors are distinguished based on the asymptotic analysis. In this section, we discuss an analytic method to evaluate the transmitted amplitude $E(z, t)$ consisting of a total transient contribution (the precursors, $E_{SB}(z, t)$) and a steady-state contribution (the main signal, $E_C(z, t)$).

By assuming that the medium is weakly dispersive ($\omega_p \ll \sqrt{8\delta\omega_0}$), has a narrow resonance ($\delta \ll \omega_0$), and that the carrier frequency of the pulse is near the material resonance ($\omega \simeq \omega_0$), the index of refraction $n(\omega)$ [Eq. (1)] is approximately given as

$$n(\omega) \simeq 1 - \frac{1}{4} \frac{\omega_p^2}{\omega(\omega - \omega_0 + i\delta)}. \quad (11)$$

With this approximation, Eq. (4) becomes

$$E(z, t) = \text{Re} \left[\frac{iE_0}{2\pi} \int_{ia-\infty}^{ia+\infty} \frac{e^{-i\omega\tau - ip/(\omega - \omega_0 + i\delta)}}{\omega - \omega_c} d\omega \right], \quad (12)$$

where $\tau = t - z/c$. Note that these assumptions are also used in the SVA approximation [35], with the additional step of assuming a slowly varying amplitude, which we do not assume here. Therefore, the assumptions used to obtain Eq. (12) do not lead to the SVA approximation, but instead describe a “weakly dispersive, narrow-resonance (WDNR) assumption”. It is possible to obtain a simple analytic solution of Eq. (12) by contour integration. This solution describes the propagation of the step-modulated field through the dielectric, as discussed in several references [13, 33, 35, 37, 40]. The total transmitted field amplitudes are given as

$$A(z, t) = \text{Re} \left[E_0 \Theta(\tau) \left(e^{\frac{p}{i\Delta - \delta}} - e^{(i\Delta - \delta)\tau} \sum_{n=1}^{\infty} \left(\frac{\sqrt{p/\tau}}{i\Delta - \delta} \right)^n J_n(2\sqrt{p\tau}) \right) \right], \quad (13)$$

$$A(z, t) = \text{Re} \left[E_0 \Theta(\tau) e^{(i\Delta - \delta)\tau} \sum_{n=0}^{\infty} \left(\frac{-i\Delta + \delta}{\sqrt{p/\tau}} \right)^n J_n(2\sqrt{p\tau}) \right], \quad (14)$$

$$A_C(z, t) = \text{Re} [E_0 \Theta(\tau) e^{\frac{p}{i\Delta - \delta}}]. \quad (15)$$

where Eq. (13) [Eq. (14)] should be used for $\tau > p/|i\Delta - \delta|^2$ [$\tau < p/|i\Delta - \delta|^2$]. The first term of Eq. (13) corresponds to the main signal $E_C(z, t) = \text{Re} [E_0 \Theta(\tau) e^{p/(i\Delta - \delta)} e^{-i\omega_c\tau}]$. The amplitude of the main signal increases as the detuning Δ increases because there is less absorption,

as shown in Fig. 5 (c). The second term is the sum of Sommerfeld and Brillouin precursors $E_{SB}(z, t)$, where the peak of the total precursor envelope decreases as the detuning increases [see Fig. 5 (b)]. The field envelope is expressed as $A(z, t)$, such that $E(z, t) = A(z, t)e^{-i\omega_c\tau}$, $E_{SB}(z, t) = A_{SB}(z, t)e^{-i\omega_c\tau}$, and $E_C(z, t) = A_C(z, t)e^{-i\omega_0\tau} = A_C(z, t)e^{i\Delta\tau}e^{-i\omega_c\tau}$. To compare this theory to our experimental data, the normalized transmitted intensity $T(z, t) = |E(z, t)/E_0|^2$ is evaluated from Eqs. (13)-(14).

For the case of an off-resonance carrier frequency ($\Delta \neq 0$), the WDNR approximation (blue dashed-lines in Fig. 3) shows that the modulation of the total transmitted intensity depends on the detuning Δ [See Fig. 3 (a), (b), (d), and (e)]. This is because the total precursor field [the second term in Eq. (13)] always oscillates at the medium resonance frequency $\omega_0 (= -\Delta + \omega_c)$, while the main signal (first term) oscillates at the carrier frequency of the incident pulse ω_c . The modulation pattern is also explained by the cross term $E_{SB}(z, t)^* E_C(z, t)$ of $T(z, t)$ as

$$T(z, t) = |E(z, t)/E_0|^2 \quad (16)$$

$$\begin{aligned} &\simeq |E_{SB}(z, t)/E_0|^2 + |E_C(z, t)/E_0|^2 \\ &+ \frac{2p}{\sqrt{\Delta^2 + \delta^2}} \frac{J_1(2\sqrt{p\tau})}{\sqrt{p\tau}} e^{-\delta\tau - \delta p/(\Delta^2 + \delta^2)} \\ &\times \cos\left(\Delta\tau + \frac{\Delta p}{\Delta^2 + \delta^2} + \varphi(\Delta)\right) + \dots, \end{aligned} \quad (17)$$

where we have retained the dominant first term ($n = 1$) and dropped the higher order terms in Eq. (13). The modulation frequency is $\Delta/2\pi$, as shown in Eq. (17). Therefore, it is confirmed that the difference in frequencies between total precursors and main signal gives rise to modulation of the total transmitted field intensity.

The modulation patterns in Fig. 3 were also observed by Hamermesh *et al.* (Fig. 6-9 of [35]) when they studied the time-dependent emission of gamma rays propagating through an absorptive filter. They detuned the energy of the gamma rays from the resonant energy of the filter. [Note that, although their analysis is based on a single-sided decaying exponential input pulse, it can be modified to a step-modulated pulse when we set the exponent to zero.] They also observed transient non-exponential decay of resonantly filtered gamma rays, which might be another realization of precursors in electromagnetic pulse propagation.

The transmitted intensity measured in our experiment is smoothed out by the finite bandwidth of our measuring system. To take into account the finite rise time of the step-

modulated pulse and detection system, we convolve the theoretically predicted intensity transmission function $T(z, t)$ with the expression for a single-pole low-pass filter

$$\frac{dy(t)}{dt} = -\gamma_f[y(t) - T(z, t)], \quad (18)$$

where $y(t)$ is the filtered transmission function and $\gamma_f = 2\pi$ (206-MHz) is the 3-dB roll-off frequency in rad/s. The filter reduces the transmission to $\sim 95\%$ immediately after the front. The blue dashed lines in Fig. 3 shows our predictions of the low-pass-filtered intensity transmission function, which agrees well with the experimental observations.

VI. COMPARISON OF TWO THEORIES

As discussed in previous sections, we observed discrepancies between two theories near $\tau = 0$ for our experimental condition of small optical depth, $\alpha_0 L \sim 1$. In this section, by increasing $\alpha_0 L$, we compare the total precursor $A_{SB}(z, t)$ for two different theories: the asymptotic method $[A_S(z, t) + A_B(z, t)]$ and the analytic method for the total precursors [Eqs. (13)-(14)]. Figure 6 shows a comparison of the two theories for different values of $\alpha_0 L$ that go beyond what we can obtain in our experiments.

For the optically thin cases [Fig. 6 (a)-(b)], the results still shows deviation near $\tau \sim 0$. For the optically mature regime ($\alpha_0 \gg L^{-1}$), however, the two theories agree well, as discussed in Ref. [25]. The oscillation due to the beating of the absorption-created sidebands, which is characterized by the Bessel functions $J_n(2\sqrt{p\tau})$ appearing in Eqs. (13)-(14), is reproduced by the analytic asymptotic theory [12, 49], but not by the previous asymptotic theory [10].

VII. DISCUSSION

We discuss here the experimental details of our previous report on the first observation of optical precursors for the case of resonant excitation of the medium ($\Delta = 0$) [11] and extend the discussion to include the case of off-resonance $\Delta \neq 0$. By finding experimental conditions that are ideal for observing long-lived optical precursors with significant amplitude, we detected essentially 100% transmission of the leading edge of the incident field propagating through the medium; after the transient, the later part of the field decays to the value

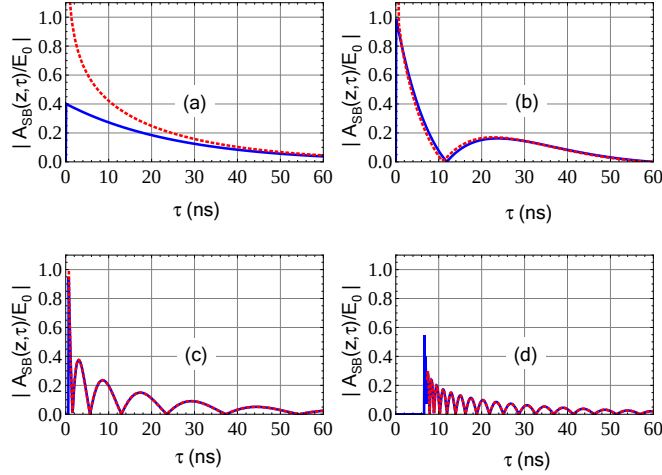


FIG. 6: The absolute value of the total transient field envelope for the weakly dispersive narrow resonance case (blue solid lines) and the asymptotic theory (red dashed lines) [49] for $\Delta = 0$. Each case has different medium distances z : (a) 0.2 cm, (b) 2 cm, (c) 20 cm, and (d) 200 cm.

expected from Beer's Law. The time scale of the precursors is inversely related to the linewidth δ . We achieve precursors with time scales measured in tens of nanoseconds using a narrow resonance of a cold potassium atomic gas. The rise time of our pulse T_r is of the order of 100 ps, which is less than the expected time scale of the precursor $1/\delta$ [p. 423 of Ref. [10]]. We use the recent theory of LeFew *et al.* Ref. [49] to identify Sommerfeld and Brillouin precursors. This theory is valid for the weakly dispersive and singular case as appropriate our experimental conditions [11].

For off-resonance carrier frequencies, the different frequencies between the precursors and the main signal cause interference between the two and generate an oscillatory modulation of the total transmitted intensity. Note that this interference will disappear as one increases the optical depth to the point where the main signal is entirely absorbed. For the optically thick medium, the optical precursors are the only part of the signal that survive in transmission and the Bessel oscillatory behavior dominates.

Comparing these two theories also precisely confirmed that Crisp's transients [13], or the coherent transients evaluated by Macke *et al.* [33], are optical precursors, as discussed by several researchers [37, 40] and more recently by Macke *et al.* [29]. Equation (12) is equivalent to Macke and Segard's expression in the time domain [33] in which the medium

response (characterized by a Green's function) is convolved with the input pulse in the time domain.

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